



Under review as a conference paper at ICAIS 2025

DEEP LEARNING-AUGMENTED SCORE MATCHING FOR HANDLING MISSING DATA

Anonymous authors
Paper under double-blind review

ABSTRACT

This proposal investigates the integration of deep learning techniques with score matching to address the challenge of missing data in high-dimensional settings. Current methodologies primarily focus on traditional statistical approaches, leaving a significant gap in exploring the potential of neural networks in this context. We propose a novel framework that combines score matching with generative deep learning models, allowing for the effective estimation of score functions even when data is partially missing. Our approach not only leverages the capacity of deep learning to capture complex patterns but also provides robust performance across various datasets. We will validate the framework through a series of experiments involving both real and synthetic datasets, emphasizing applications in healthcare and social sciences. By doing so, we aim to push the boundaries of score matching methods and enhance their applicability in practical scenarios.

1 INTRODUCTION

Handling missing data is a critical issue in many real-world applications, particularly in healthcare and social sciences. Traditional methods for addressing missing data often rely on imputation techniques that fail to consider the underlying structure of the data. Score matching, proposed by Hyvärinen (Hyvärinen, 2005), offers a promising approach to learn from incomplete data by matching the score functions. However, existing adaptations, such as those discussed by Givoni et al. (Givoni et al., 2023), have not thoroughly explored the potential of deep learning techniques. Our work aims to bridge this gap by integrating deep learning with score matching to improve robustness and performance in the presence of missing data.

2 RELATED WORK

Several studies have explored the challenges associated with missing data. For instance, Park et al. (Park et al., 2022) demonstrated the efficacy of deep learning approaches for imputing values in time series data, while Zhou (Zhou, 2020) focused on convolutional neural networks for robust prediction under missing conditions. However, these methods do not directly integrate score matching principles. Our work addresses this oversight by combining deep learning frameworks with score matching techniques, enhancing the estimation of score functions in situations where data is incomplete.

3 METHOD

We propose a deep learning framework that integrates variational autoencoders (VAEs) (Kingma & Welling, 2013) with score matching techniques. The model learns to estimate the score function of the data distribution while effectively handling missing entries. The framework consists of a VAE architecture where missing data points are imputed during the training phase, allowing the model to learn from both complete and incomplete data. This dual approach not only improves the model's robustness but also helps it to generalize better across various datasets.

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4 EXPERIMENTAL SETUP

Our experiments focus on evaluating the proposed framework on synthetic datasets generated with controlled missingness patterns, as well as real-world datasets from healthcare (Burns et al., 2024) and social sciences. We compare our method against traditional score matching techniques and

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 **Meta Review Status:** The meta review and decision have not been released yet.

AI Review from DeepReviewer

 AI Review from DeepReviewer will be automatically processed

 Summary

This paper introduces a novel framework that aims to integrate deep learning techniques with score matching to address the challenge of handling missing data in high-dimensional settings. The authors propose leveraging the power of generative deep learning models, specifically Variational Autoencoders (VAEs), to estimate score functions even when data is incomplete. The core idea is to combine the ability of deep learning to capture complex data patterns with the robustness of score matching, which is a technique used to estimate the gradient of the log-density of a probability distribution. The authors argue that traditional statistical methods for handling missing data often fall short in capturing these complex patterns, and that their proposed framework offers a more effective solution. The methodology involves training a VAE where missing data points are imputed during the training phase, allowing the model to learn from both complete and incomplete data. The score function is then estimated using the learned VAE representation. The paper presents experimental results on both synthetic and real-world datasets, including data from healthcare and social sciences, to demonstrate the effectiveness of the proposed framework. The experiments include evaluations of the impact of varying missing data patterns and the use of dropout regularization. The authors claim that their approach not only leverages the capacity of deep learning to capture complex patterns but also provides robust performance across various datasets. However, the paper lacks a detailed explanation of the specific deep learning architecture used, the loss function, and the optimization algorithm. Furthermore, the experimental evaluation is limited in scope, and the paper does not provide a thorough discussion of the limitations of the proposed method. Despite these shortcomings, the paper presents a potentially valuable contribution to the field of missing data imputation by exploring the integration of deep learning and score matching. The overall significance of the work is currently limited by the lack of methodological detail and the insufficient experimental validation, but the core idea is promising and warrants further investigation.

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 Strengths

The primary strength of this paper lies in its conceptual contribution: the proposal to integrate deep learning techniques, specifically Variational Autoencoders (VAEs), with score matching to address the challenge of missing data in high-dimensional settings. This is a novel approach that has the potential to combine the strengths of both methodologies. Deep learning models, particularly VAEs, are well-suited for capturing complex, non-linear patterns in data, while score matching is a robust method for estimating the gradient of the log-density of a probability distribution, which is crucial for handling missing data. The authors correctly identify a gap in traditional statistical methods, which often struggle to capture such complex patterns. The idea of imputing missing data during the training phase of the VAE is also a notable contribution, as it allows the model to learn from both complete and incomplete data, potentially leading to a more robust and accurate estimation of the score function. Furthermore, the authors' decision to validate their framework on both synthetic and real-world datasets, including data from healthcare and social sciences, is commendable. This demonstrates an awareness of the practical relevance of their work and a desire to evaluate its performance in realistic scenarios. The inclusion of experiments that investigate the impact of varying missing data patterns is also a positive aspect, as it acknowledges the complexity of real-world missing data scenarios. The authors also explore the impact of dropout regularization, which is a common technique used to prevent overfitting in deep learning models. Finally, the paper is generally well-structured and clearly written, making it easy to follow the proposed method and its validation. While the paper has significant weaknesses, the core idea of combining deep learning and score matching for missing data imputation is a promising avenue for future research.

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 Weaknesses

After a thorough review of the paper and the provided analyses, I have identified several significant weaknesses that undermine the validity and impact of the work. These weaknesses primarily revolve around the lack of methodological detail, the limited scope of the experimental evaluation, and the absence of a thorough discussion of the proposed method's limitations. Firstly, the paper suffers from a severe lack of methodological detail, making it difficult to understand the proposed

framework and reproduce the results. The authors state that they integrate VAEs with score matching, but they fail to provide specific details on how this integration is achieved. The paper mentions that the model learns to estimate the score function while handling missing entries (Section 3), but it does not explain how score matching is adapted to handle missing data. The description of the VAE architecture is also insufficient. The authors do not specify the encoder and decoder architectures, the dimensionality of the latent space, or the activation functions used. The loss function and optimization algorithm are not explicitly stated, and there are no mathematical derivations or algorithmic steps provided. This lack of detail is a major impediment to understanding the proposed method and assessing its novelty and effectiveness. The absence of a clear explanation of how the imputation is performed within the training loop further compounds this issue. This lack of clarity is a significant barrier to both understanding and replicating the work, and I have a high level of confidence in this assessment. Secondly, the experimental evaluation is not comprehensive enough to support the claims made by the authors. The paper mentions using synthetic datasets with controlled missingness patterns and real-world datasets from healthcare and social sciences (Section 4), but it does not provide specific details about these datasets. The specific real-world datasets used are not named, and the paper lacks details on the number of samples, the number of features, and the type of data. The paper states that the method is compared against traditional score matching techniques and benchmark imputation methods (Section 4), but it does not specify which methods are used for comparison. This lack of detail makes it difficult to assess the relative performance of the proposed method. Furthermore, the paper does not provide a justification for the choice of evaluation metrics, which are mean squared error and reconstruction error. The results section primarily presents loss curves without statistical analysis, such as confidence intervals or p-values. The impact of different missingness mechanisms (MCAR, MAR, MNAR) is not thoroughly investigated, despite the mention of an experiment on varying missing data patterns. The lack of statistical analysis and the limited number of datasets used make it difficult to draw any firm conclusions about the effectiveness of the proposed method. I have a high level of confidence in this assessment, as the lack of detail in the experimental setup and the absence of statistical analysis are clearly evident in the paper. Finally, the paper lacks a thorough discussion of the limitations of the proposed method. The authors do not discuss the computational complexity of the method, its sensitivity to hyperparameters, or its performance under different missingness mechanisms. The paper also fails to address the potential challenges in applying the method to real-world datasets, such as the presence of outliers or noisy data. The authors do not discuss the potential for the method to introduce bias into the analysis, nor do they provide guidance on how to mitigate this bias. The absence of a dedicated limitations section and the lack of discussion on these critical aspects significantly undermine the practical relevance of the work. The paper also does not discuss the scalability of the method to large datasets, which is a crucial consideration for real-world applications. I have a high level of confidence in this assessment, as the paper clearly lacks any discussion of these important limitations. In summary, the paper's lack of methodological detail, limited experimental evaluation, and absence of a thorough discussion of limitations are significant weaknesses that need to be addressed before the work can be considered a valuable contribution to the field.

 Suggestions

To address the identified weaknesses, I recommend several concrete and actionable improvements. Firstly, the authors must provide a detailed explanation of their proposed framework. This should include a clear description of the VAE architecture, specifying the number of layers, the type of activation functions, and the dimensionality of the latent space. The authors should also provide the mathematical equations for the encoder and decoder networks, as well as the loss function used for training. The specific score matching algorithm employed should be detailed, including the loss function and the optimization procedure. Furthermore, the authors need to clarify how the imputation of missing data is performed within the training loop and how the score matching is adapted to handle incomplete data. A pseudocode of the algorithm would greatly enhance the clarity and reproducibility of the method. The authors should also provide a detailed explanation of how the proposed method addresses the limitations of existing methods, highlighting the specific advantages of their approach. Additionally, the authors should discuss the computational complexity of the method, its sensitivity to hyperparameters, and its performance under different missingness mechanisms. Secondly, the experimental evaluation needs to be significantly expanded to provide a more comprehensive assessment of the proposed method. This should include comparisons with a wider range of existing methods for handling missing data, including both imputation-based methods and methods that directly handle missing data during the learning process. The comparison should be performed on a diverse set of datasets with varying characteristics, including different data types (e.g., continuous, categorical), different missingness patterns (e.g., missing completely at random, missing at random, missing not at random), and different dimensionality. The authors should also provide a detailed analysis of the performance of their method under different missingness scenarios, and they should use appropriate evaluation metrics, such as mean squared error, root mean squared error, and R-squared, to assess the quality of the imputations. Furthermore, the authors should investigate the sensitivity of their method to different hyperparameter settings and provide guidelines for selecting appropriate values. The experimental setup should be described in more detail, including the size of the datasets, the number of missing values, and the specific missingness patterns used. The authors should also include a discussion of the computational cost of the method, including the training time and memory usage. Finally, the authors should provide a more thorough discussion of the limitations of their method. This should include a discussion of the computational complexity of the method, its sensitivity to hyperparameters, and its performance under different missingness mechanisms. The authors should also discuss the potential challenges in applying the method to real-world datasets, such as the presence of outliers and noise. They should also discuss the potential for the method to introduce bias into the analysis, and they should provide guidance on how to mitigate this bias. A discussion of the scalability of the method to large datasets is also needed, including an analysis of the memory and time requirements of the algorithm. The authors should also discuss the potential for the method to be extended to other types of data, such as time series data or image data. By addressing these points, the authors can significantly improve the quality and impact of their work.

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 Questions

Based on my analysis, I have several questions that I believe are crucial for clarifying the methodology and understanding the limitations of the proposed framework. Firstly, how does the proposed framework specifically handle different types of missing data patterns, such as missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR)? The paper mentions an experiment on varying missing data patterns, but it does not provide a detailed analysis of how the method performs under each of these scenarios. Understanding the method's behavior under different missingness mechanisms is crucial for assessing its applicability to real-world datasets. Secondly, what are the computational requirements of the proposed framework, and how does it scale to large datasets? The paper does not provide any analysis of the time and space complexity of the algorithm, which is a critical consideration for practical applications. Understanding the computational cost and scalability of the method is essential for determining its feasibility for real-world use. Thirdly, how does the proposed framework compare to other deep learning-based methods for handling missing data, such as those based on generative adversarial networks (GANs) or other variants of VAEs? The paper mentions comparisons with traditional score matching techniques and benchmark imputation methods, but it does not provide a detailed comparison with other deep learning-based approaches. A thorough comparison with state-of-the-art deep learning methods is necessary to assess the relative advantages and disadvantages of the proposed framework. Finally, can the proposed framework be extended to other types of data, such as time series data or image data? The paper focuses on tabular data, but it is unclear whether the method can be readily applied to other types of data. Understanding the generalizability of the method to different data types is important for assessing its broader applicability. These questions target core methodological choices and assumptions, and answering them would significantly enhance the understanding and impact of the work.

 **Scores**

Soundness: 1.5

Presentation: 1.5

Contribution: 1.5

Rating: 3.5

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AI Review from ZGCA



 **Summary**

The paper proposes a framework that integrates score matching with a deep generative model (a Variational Autoencoder, VAE) to handle missing data in high-dimensional settings. The central idea is to 'actively impute' missing entries during VAE training so that the model can learn from both complete and incomplete observations while estimating the score function of the data distribution. The authors evaluate on synthetic and real datasets (including a healthcare dataset) and report training/validation loss curves under different hidden layer sizes, dropout rates, and missingness mechanisms (MCAR, MAR, NMAR), claiming that the approach is robust and that a hidden size of 128 performs best.

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 **Strengths**

- Addresses an important, widely applicable problem: robust learning with missing data in high-dimensional settings.
- Conceptual integration of score matching with a deep generative model (VAE) is promising and could leverage the representational power of neural networks for estimating scores under missingness (Section 3).

- Considers multiple missingness types (MCAR/MAR/NMAR) and regularization settings (dropout), which are relevant factors when evaluating robustness (Section 5).
- Preliminary analysis of hyperparameters (hidden layer sizes, dropout) and training dynamics (loss curves) provides an initial view of model behavior.

✗ Weaknesses

- Insufficient methodological detail: the paper does not specify the precise score matching objective (e.g., denoising or Hyvärinen score), how missing entries are handled in the score estimation (masking/marginalization), or how the VAE and score estimator are coupled during training. Section 3 remains high-level with no equations, pseudo-code, or algorithmic steps.
- Claims of robustness and improved performance are not substantiated by rigorous experiments: baselines are described vaguely as 'traditional score matching techniques and benchmark imputation methods' (Section 4) without naming or configuring them, and no quantitative comparisons (e.g., MSE, MAE, NLL) with confidence intervals are reported.
- Experimental reporting is limited to loss curves; there are no tables or quantitative metrics comparing against established methods (e.g., GAIN, missForest, MICE, kNN, RF imputation, diffusion/score-based imputation), hindering assessment of effectiveness.
- Reproducibility gaps: missing dataset details (sizes, features, variable types), missingness generation protocols (percentages, per-feature patterns), seeds, hardware, training schedules, and optimization details beyond hidden layer sizes. No code or resource statement.
- Handling of different missingness mechanisms is asserted but not formally modeled: no explicit treatment of MAR or NMAR (e.g., likelihood factoring, missingness model), yet robustness to NMAR is claimed via loss curves (Section 5).
- Related work is underdeveloped: it cites VAEs and score matching but does not situate the approach against recent score-based/diffusion methods used for imputation or hybrid deep imputation frameworks; one citation is incomplete ('Givens et al.').
- No theoretical analysis or guarantees (e.g., consistency of score estimators under missingness, identifiability, or bias under MAR/NMAR), and no ablations that isolate the effect of the proposed coupling versus standard VAE imputation.
- Clarity issues: key concepts like 'active imputation during training' are described at a high level without operational definitions, leaving room for multiple interpretations.

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? Questions

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- Please formalize the learning objective: What exact score matching variant is used (Hyvärinen, denoising score matching, sliced score matching)? Please specify the full objective, including how missing entries are incorporated (e.g., masked gradients, conditional score over observed variables, or marginalization).
- How is 'active imputation during training' operationalized? Is there a masking mechanism in the encoder/decoder, a learned conditional prior, or a joint loss that couples imputation error with score estimation? Please provide pseudo-code and module-level details (I/O shapes, training loop, optimization).
- What missingness mechanisms are assumed during training and evaluation? How are MAR and NMAR modeled (e.g., an explicit missingness model $p(M|X)$, inverse propensity weighting, or variational treatment)?
- Which baselines were used exactly? Please list and configure them (e.g., mean/median, kNN, missForest, MICE, GAIN, diffusion/score-based imputation, standard VAE imputation), including hyperparameters, tuning protocol, and references.
- What datasets were used (names, sizes, feature types, train/val/test splits)? For synthetic data, specify the data-generating processes and missingness rates/patterns; for real data (e.g., the NIHR dataset), describe preprocessing and the nature/extent of missingness.
- Please report quantitative metrics beyond loss curves (e.g., RMSE/MAE on held-out ground-truth for synthetic data, downstream predictive performance, NLL), averaged over multiple seeds with confidence intervals.
- How does the proposed method compare in computational cost (training time, memory, wall-clock) to established baselines? Include hardware specs and seeds for reproducibility.
- Can you provide an ablation isolating the contribution of score matching vs. a vanilla VAE imputation pipeline? Does the proposed coupling improve over standard VAE training with masking or denoising objectives?
- What failure modes do you observe? Please include qualitative examples of imputed entries, error breakdowns by feature type and missingness level, and analysis of NMAR settings.

- Will you release code and scripts to reproduce the experiments?

⚠ Limitations

- The approach may be sensitive to the VAE's inductive biases and capacity; mis-specified latent structure can yield biased scores and imputations, especially under MAR/NMAR.
- Robust handling of NMAR typically requires explicit modeling of the missingness process; without it, imputations may be biased and harm downstream analyses.
- Computational overhead of jointly learning a VAE and score estimator could be substantial for high-dimensional data; scalability to very large or mixed-type datasets (categorical, count) may be limited.
- Potential societal impacts include propagating or amplifying bias when imputing clinical or social variables, potentially affecting downstream decisions if uncertainty is not quantified and communicated.
- Generalization to time series or relational data is non-trivial; additional structural modeling may be needed.

🖼 Image Evaluation

Cross-Modal Consistency: 20/50

Textual Logical Soundness: 12/30

Visual Aesthetics & Clarity: 10/20

Overall Score: 42/100

Detailed Evaluation (≤500 words):

Visual ground truth (Figure-alone digest)

- Figure 1/(a): Training loss vs epochs for hidden sizes {32,64,128}; lines show monotonic decrease; 128 drops fastest.
- Figure 1/(b): Validation loss vs epochs; same sizes; 128 lowest.
- Figure 1/(c): Loss with BatchNorm; train/val nearly overlapping; spikes around epochs ≈50,100,150.
- Figure 1/(d): Loss without BatchNorm; similar pattern, slightly lower peaks.
- Figure 2/(a–c): Dropout panels for rates 0.0, 0.2, 0.5; each displays “No Data ...” on empty axes.
- Figure 2/(d–f): MCAR/MAR/NMAR loss curves; train/val overlap; converge near zero by ≈10 epochs.

Synopsis: Fig.1 compares hidden size and BatchNorm effects; Fig.2 intends to show dropout and missingness-pattern robustness.

1. Cross-Modal Consistency

- Major 1: Figure 1 orientation mismatch (text says left/right; caption says up/down). Evidence: “The left subplot shows... right subplot...” (Sec 5) vs “Figure 1: Up... Down...”.
- Major 2: Unresolved citation “Figures 2 and ??”. Evidence: “Figures 2 and ??” (Sec 5).
- Major 3: Claim about dropout 0.2 lacks supporting curves. Evidence: “rate of 0.2 significantly mitigates overfitting” (Sec 5) vs Fig. 2(a–c) “No Data”.
- Minor 1: Multiple unlabeled/duplicated images surrounding Fig. 1/2 create ordering ambiguity. Evidence: extra images before/after “Figure 1/2” labels.
- Minor 2: BatchNorm section title vs caption order ambiguous. Evidence: “Impact of Batch Normalization on Model Training” appears both before and after Fig. 1 label.

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2. Text Logic

- Major 1: Comparative claims lack quantitative results vs baselines. Evidence: “We compare our method against traditional score matching techniques...” (Sec 4); no tables/metrics are shown.
- Major 2: Method lacks precise score-matching formulation or loss; unclear how VAE integrates with score estimation. Evidence: “integrates VAEs with score matching... learns to estimate the score function” (Sec 3).
- Minor 1: Incomplete citation. Evidence: “Givens et al. ()” (Sec 1).
- Minor 2: Broad claims of robustness without statistical testing. Evidence: “illustrates the framework’s robustness” (Sec 5).

3. Figure Quality

- Major 1: Dropout panels are placeholders (“No Data”), blocking interpretation. Evidence: Fig. 2(a–c) text “No Data for Dropout Rate...”.
- Minor 1: Some fonts small but readable; axes lack units and dataset identifiers. Evidence: y-axis always “Loss”; no dataset names visible.
- Minor 2: Spikes in BatchNorm plots lack annotations explaining restarts. Evidence: Fig. 1(c,d) sharp peaks near 50/100/150 epochs.

Key strengths:

- Clear hidden-size comparison; legends/axes readable in Fig. 1(a,b).
- Topic is timely; integrating score matching with deep generative models is promising.

Key weaknesses:

- Severe figure–text mismatches and placeholder panels.
- Missing methodological specifics and absent baseline comparisons.
- Ambiguous figure ordering and captions; unresolved “??”.

Actionable suggestions:

- Provide complete, consistently labeled multi-pane figures with (a–f) labels.
- Add tables/plots comparing against score-matching and imputation baselines with metrics and CIs.
- Specify the score-matching objective, handling of missingness (MCAR/MAR/NMAR), and training details.
- Fix citations and replace placeholder dropout panels with real results.

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 Scores

Originality: 2

Quality: 1

Clarity: 2

Significance: 2

Soundness: 1

Presentation: 2

Contribution: 2

Rating: 3

Summary

This paper proposes a method for handling missing data by integrating score matching with variational autoencoders (VAEs). The core idea is to leverage the generative capabilities of VAEs to impute missing values during training, allowing the model to learn from both complete and incomplete data. The authors claim that this approach enhances the robustness of score matching in high-dimensional settings. The method is evaluated on synthetic datasets with controlled missingness patterns, as well as real-world datasets from healthcare and social sciences. The experiments explore the impact of hyperparameter tuning, batch normalization, and dropout regularization on the model's performance. The primary metrics used for evaluation are mean squared error (MSE) and reconstruction error. The authors compare their method against traditional score matching techniques and benchmark imputation methods, although the specific details of these comparisons are not fully elaborated in the provided text. The paper aims to address the limitations of traditional imputation techniques by providing a more robust and flexible approach that can handle complex data patterns. While the integration of score matching and VAEs is presented as a novel contribution, the specific mechanisms of this integration and the novelty of the approach are not clearly articulated. The experimental results, while demonstrating the effectiveness of the proposed method, lack detailed numerical comparisons with baseline methods, making it difficult to fully assess the magnitude of the improvement. Overall, the paper presents an interesting approach to handling missing data, but it suffers from a lack of clarity in the method description, insufficient experimental details, and a weak theoretical justification for the proposed method's advantages over existing techniques. The paper also lacks a thorough discussion of the limitations of the proposed approach and its potential for generalization to other types of data and missingness mechanisms. The writing quality is also a significant issue, with numerous grammatical errors and typos throughout the text.

Strengths

The core strength of this paper lies in its attempt to address the pervasive problem of missing data in high-dimensional settings by integrating deep learning techniques, specifically variational autoencoders (VAEs). The idea of using a VAE to impute missing values during training is a novel approach, as it allows the model to learn from both complete and incomplete data, which could lead to a more robust representation of the data distribution. The authors' recognition of the limitations of traditional imputation techniques is also a valid point, as these methods often fail to capture the complex patterns present in high-dimensional data. Furthermore, the paper explores the impact of various hyperparameters, such as hidden layer size and dropout, on the model's performance. This exploration is important for understanding the practical aspects of implementing the proposed method and optimizing its performance. The use of both synthetic and real-world datasets for evaluation is also a positive aspect, as it provides a more comprehensive assessment of the method's effectiveness. The inclusion of real-world datasets from healthcare and social sciences highlights the potential practical applications of the proposed method. While the paper has significant weaknesses, the core idea of combining score matching with VAEs for handling missing data is a promising direction for future research. The authors' attempt to move beyond traditional imputation methods and leverage the power of deep learning is a valuable contribution. The experimental results, despite lacking detailed numerical comparisons, do suggest that the proposed method can achieve good performance under certain hyperparameter configurations. The exploration of different missing data patterns (MCAR, MAR, NMAR) is also a strength, as it demonstrates the method's robustness under various conditions.

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Weaknesses

After a thorough review of the provided paper, I have identified several significant weaknesses that undermine its overall quality and impact. Firstly, the **method description lacks clarity and detail**. While the paper mentions the use of VAEs and score matching, it does not provide a concrete mathematical formulation of how these two components are integrated. The paper states that the VAE imputes missing data during training, but it does not explain the specific mechanisms of this imputation process or how the score matching objective is incorporated into the VAE training. This lack of detail makes it difficult to understand the proposed method fully and to reproduce the results. The description of the model architecture is also vague, lacking specifics on the number of layers, activation functions, and the dimensionality of the latent space. This lack of clarity is a major obstacle to understanding the paper's core contribution. My confidence in this assessment is high, as the paper consistently lacks specific mathematical formulations and algorithmic details. Secondly, the **experimental section lacks crucial details and numerical results**. While the paper mentions comparisons with traditional score matching techniques and benchmark imputation methods, it does not specify which methods were used as baselines. The results are primarily presented as loss curves, which are qualitative, and the paper does not provide a table with numerical results (e.g., MSE values) for different methods and datasets. This lack of numerical results makes it impossible to quantitatively assess the performance of the proposed method and to compare it with existing approaches. The paper also lacks details on the synthetic data generation process, such as the underlying data distribution and the

method of introducing missingness. Similarly, the real-world datasets are mentioned, but their specific characteristics and preprocessing steps are not described. This lack of detail makes it difficult to interpret the results and to assess the generalizability of the proposed method. My confidence in this assessment is high, as the paper explicitly lacks numerical results and detailed dataset descriptions. Thirdly, the paper suffers from a **weak theoretical justification**. The paper claims that the proposed method is more robust than traditional imputation techniques, but it does not provide a theoretical analysis or proof to support this claim. The paper does not explain why the specific combination of VAEs and score matching is advantageous for handling missing data compared to other existing methods. The paper also lacks a discussion of the limitations of the proposed approach and its potential for generalization to other types of data and missingness mechanisms. This lack of theoretical grounding weakens the paper's overall contribution. My confidence in this assessment is high, as the paper lacks any formal theoretical analysis or proofs. Fourthly, the paper has **significant writing and formatting issues**. There are numerous grammatical errors, typos, and inconsistencies in formatting throughout the text. The missing figure reference in the experiments section is a clear indication of these issues. These issues make the paper difficult to read and understand, further hindering its impact. My confidence in this assessment is high, as these issues are directly observable in the provided text. Finally, the paper **lacks novelty**. While the paper proposes a combination of VAEs and score matching, this combination is not entirely novel, as similar approaches have been explored in the literature. The paper does not adequately discuss or differentiate its approach from existing methods, making it difficult to assess its unique contribution. My confidence in this assessment is medium, as the paper does not provide sufficient evidence to support the novelty of its approach. In summary, the paper's weaknesses stem from a lack of clarity in the method description, insufficient experimental details, a weak theoretical justification, significant writing issues, and a lack of novelty. These weaknesses significantly undermine the paper's overall quality and impact.

 Suggestions

To address the identified weaknesses, I recommend the following concrete and actionable improvements. Firstly, the authors should provide a **detailed and clear explanation of the proposed method**. This should include a mathematical formulation of how score matching is integrated with the VAE architecture, specifying the objective function and the training procedure. The authors should also provide details on the VAE architecture, including the number of layers, activation functions, and the dimensionality of the latent space. A diagram or pseudocode would be helpful to illustrate the method. The authors should also clarify how the imputation is performed during training and how the score function is estimated in the presence of missing data. This detailed explanation will address the lack of clarity and allow other researchers to understand and reproduce the results. Secondly, the authors should provide **detailed experimental results, including comparisons with baseline methods**. This should include a table with numerical results (e.g., MSE values) for different methods and datasets. The authors should also provide details on the synthetic data generation process, such as the underlying data distribution and the method of introducing missingness. The datasets should be described in detail, including their characteristics and preprocessing steps. The authors should also specify which baseline methods were used for comparison. This will allow for a quantitative assessment of the proposed method's performance and its comparison with existing approaches. Thirdly, the authors should provide a **theoretical justification for the proposed method**. This should include a discussion of why the specific combination of VAEs and score matching is advantageous for handling missing data compared to other existing methods. The authors should also provide a theoretical analysis or proof to support their claims about the method's robustness. This will strengthen the paper's overall contribution and provide a deeper understanding of the method's properties. Fourthly, the authors should **carefully proofread and edit the paper** to correct all grammatical errors, typos, and inconsistencies in formatting. This will improve the readability and overall quality. The authors should also ensure that all figures and tables are properly referenced and labeled. This will make the paper more professional and easier to understand. Fifthly, the authors should **address the novelty concerns** by clearly articulating the unique contributions of the proposed method and by comparing it with existing methods in the literature. The authors should also discuss the limitations of their approach and its potential for generalization to other types of data and missingness mechanisms. This will help to clarify the paper's contribution and its potential impact on the field. Finally, the authors should consider **expanding the experimental evaluation** to include more diverse datasets and missingness scenarios. This will provide a more comprehensive assessment of the method's effectiveness and its generalizability. By addressing these weaknesses, the authors can significantly improve the quality and impact of their paper.

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 Questions

Based on my review, I have several questions that I believe are crucial for understanding the paper's methodology and contributions. Firstly, could the authors provide a more detailed explanation of how the score matching objective is incorporated into the VAE training process? Specifically, how is the score function estimated, and how does the imputation of missing values during training affect this estimation? A mathematical formulation of this integration would be highly beneficial. Secondly, what is the specific mechanism for imputing missing values during the VAE training? Are the missing values initialized with zeros or some other value, and how are they updated during the training process? How does this imputation mechanism interact with the score matching objective? Thirdly, what is the rationale behind the choice of VAEs for this task? Are there other generative models that could be used, and what are the advantages and disadvantages of using VAEs compared to these alternatives? Fourthly, could the authors provide more details on the synthetic data generation process? What is the underlying data distribution, and how is missingness introduced? What are the characteristics of the real-world datasets used in the experiments, and what preprocessing steps were

applied? Fifthly, what are the specific benchmark imputation methods that the proposed method is compared against? Why were these specific methods chosen, and how do they relate to the proposed approach? Sixthly, what are the limitations of the proposed approach, and under what conditions might it fail to perform well? How does the method handle different missingness mechanisms (MCAR, MAR, NMAR), and what are the potential biases that could be introduced? Finally, what are the potential avenues for future research based on this work? How can the proposed method be extended or improved, and what are the most promising directions for future exploration? These questions are aimed at clarifying the core methodological choices, addressing the identified weaknesses, and exploring the potential for future research.

 Scores

Soundness: 1.0

Presentation: 1.0

Contribution: 1.0

Confidence: 4.0

Rating: 1.5

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